```
    def __init__(self):
        super(SimpleNN, self).__init__()
        self.fc = nn.Linear(2, 1)

    def forward(self, x):
        return torch.sigmoid(self.fc(x))
model = SimpleNN()
```

Key features:

- Dynamic graph creation for flexible model building.
- Native support for GPU acceleration.
- Strong community support and extensive resources.

Use cases:

- Reinforcement learning
- Generative adversarial networks (GANs)

3 Scikit-learn

Scikit-learn is a Python library for traditional machine learning algorithms. It's built on top of NumPy, SciPy, and Matplotlib.

Example: Training a simple decision tree classifier:

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier

data = load_iris()
model = DecisionTreeClassifier()
model.fit(data.data, data.target)
```

Key features:

- Wide range of supervised and unsupervised learning algorithms.
- Easy integration with other Python libraries.
- User-friendly API for rapid prototyping.

Use cases:

- Classification and regression tasks
- Dimensionality reduction

4 Keras

Keras is a high-level neural networks API written in Python, capable of running on top of TensorFlow, CNTK, or Theano.

Example: Building a neural network with Keras:

```
from keras.models import Sequential
from keras.layers import Dense

model = Sequential([
    Dense(32, activation='relu', input_shape=(784,)),
    Dense(10, activation='softmax')
])
```

Key features:

- Simple and modular architecture.
- Quick prototyping with minimal code.
- Pre-trained models available for transfer learning.

Use cases:

- Building neural networks
- Transfer learning for specific tasks

5 Apache MXNet

MXNet is a deep learning framework known for its efficiency and scalability. It's particularly useful for deploying models on the cloud.

Key features:

- Flexible programming in multiple languages (Python, Scala, R, etc).
- Optimised for both GPUs and CPUs.
- Hybrid front-end for ease of use and high performance.

Use cases:

- Speech recognition
- Autonomous driving systems

6 H2O.ai

H2O.ai offers an open source platform for data science and machine learning, making it easy to build models for business applications.

Example: Running an AutoML model:

```
import h2o
from h2o.automl import H2OAutoML

h2o.init()

data = h2o.import_file('data.csv')
model = H2OAutoML(max_models=20)
model.train(y='target', training_frame=data)
```

Key features:

- AutoML functionality for automated model selection and tuning.
- Integration with Spark and Hadoop.
- Highly scalable for large datasets.

Use cases:

- Fraud detection
- Predictive analytics

7 OpenCV

OpenCV (Open Source Computer Vision Library) is a library aimed at real-time computer vision.

Example: Reading and displaying an image: